

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited.
Document Title	Process Datasheets
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PSE-DSH-001
Document Revision	A01
Document Status	Issued for Approval
Originator	S. Bello
Security Classification	Restricted
Issue Date	23 January 2023

Process Datasheets

Emergency Shutdown Valve Datasheet | Pig Launcher Receiver Datasheet
PSV Datasheet | RO Datasheet

A01	23.01.23	Issued for Approval	S.Bello	O.Joseph	A.Duru	
R01	25.07.22	Issued for Review	S.Bello	O.Joseph	K.Yusuff	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R00	25.07.22	Issued for Interdisciplinary Checks
R01	28.07.22	Issued for Review
A01	23.01.23	Approved for Design

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

This document is made available subject to the condition that the recipient will neither use nor disclose the contents except as agreed in writing with the copyright owner. Copyright is vested in Brass Fertilizer and Petrochemical Company Limited (BFPCL) © All rights reserved.

Neither the whole nor any part of this document may be reproduced or distributed in any form or by any means (electronic, mechanical, reprographic, recording, or otherwise) without the prior written consent of the copyright owner.

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PSE-DSH-001PSE-RPT-011

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001_R00	PIPESIM Hydraulics Analysis and Report (Steady State)
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-001	Process Flow Scheme (PFS)
EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003 (sheet 1 to 8)	Process Engineering Flow Scheme (PEFS)
EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Pipe Wall Thickness Calculation Report
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-014	Process Heat and Material Balance
	FISAS ESIA Draft Report (November 2020)
EA-BFPCL-GPMC-GGPL-FEED-MPP-LST-010-R01	Valve List
EA-BFPCL/BFPP GAS PL/FEED-PSE-005B	Line List

Table of Contents

1.	INTRODUCTION	7
1.1.	Background Information	7
1.2.	Purpose	8
2.	REGULATIONS, CODES AND STANDARDS	8
3.	DATASHEETS	9
3.1.	PIG Launcher and Receiver Datasheets	10
3.2.	Emergence Shutdown Valves and Blowdown Valves Datasheets	16
3.3.	PSV Datasheets	26
3.4.	RO Datasheets	35

Abbreviations

Abbreviation	Meaning
API	American Petroleum Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society of Testing and Materials
BARG	Bar Gauge
BOD	Basis of Design
BPD	Barrel Per Day
BCPD	Barrel of Condensate Per Day
BWPD	Barrel of Water Per Day
EPCI	Engineering, Procurement, Construction, Installation
FEED	Front End Engineering Design
MMSCFD	Million Standard Cubic Feet Per Day
NPS	Nominal Pipe Size
OD	Outside Diameter
WT	Wall Thickness

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

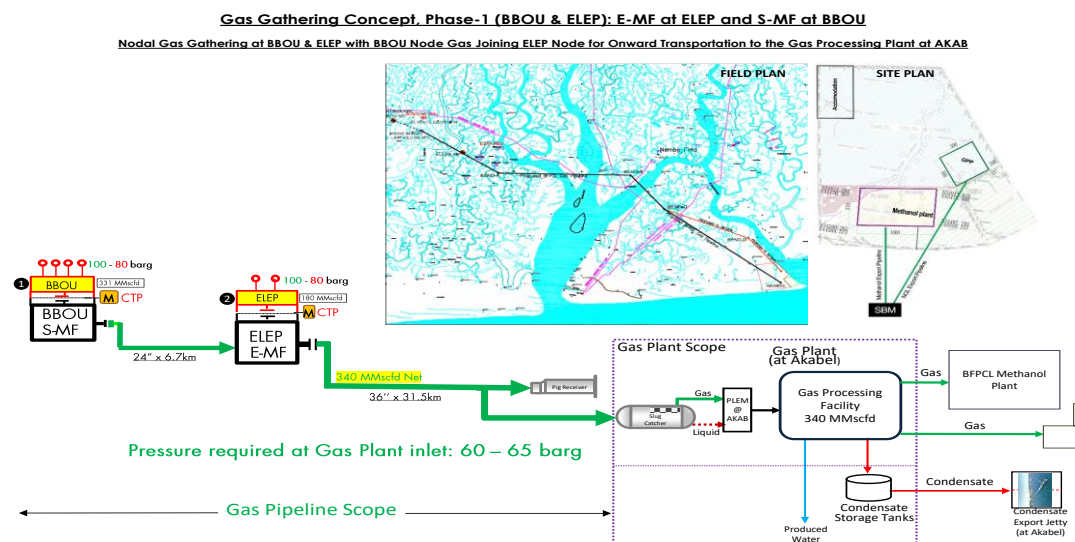


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is to clearly document the datasheets of the key equipment and components required for the successful delivery of the gas from different fields to AKABEL.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 31.40.10.13-Gen	Design of Pipeline Pig Trap Systems
API RECOMMENDED PRACTICE 14E	Recommended Practice for Design and Installation of Offshore Production Platform Piping Systems
ASME B31.8	Gas Transmission and Distribution Piping Systems
DEP 31.40.10.13-Gen	Design of Pipeline Pig Trap Systems

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Datasheets

This section summarizes the process datasheets for different equipment and items clearly specifying the process condition from different fields.

The datasheets are as in the subsections.

3.1. PIG Launcher and Receiver Datasheets

Process Data Sheet for BBOU Pig Receiver (BBOU-A-100)										
1	Tag No. : BBOU-A-100				Client: BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD					
2	Service: GAS				Plant Location: BBOU					
3	P & ID No. : EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_1				Manufacturer:					
4	Calculation Ref.: R02				Purchase Order No. :					
5	Line No.:				No. of Units: ONE					
6	Equipment No.:				Service Description.:					
11										
12	Applicable Piping Class				900#					
13	Design Code				ANSI B31.4		ANSI B31.8	☒	ANSI B31.3	☒
14	Tag Number(s)				BBOU-A-100					
15	Pipeline Diameter				24 inches (24"/28" minor/major barrels)					
16	Type of Pigs				Normal	☒	Spheres	☒	Other (state below)	
17					Intelligent	☒	Foam	☒		
18	Chemical Injection				Yes		No	☒		
19	Thermowell installation				Yes		No	☒		
20	Valve Type				Globe	☒	Ball	☒		
21	Intended Pigging Frequency				Quarterly					
22	Power-actuated Valves				Yes		No	☒		
23	Pre-commissioning Drying Method				Vacuum		Methanol		N/A	
24	Type of Valve to be used on Mainline Parts				Weld end		Flanged	☒		
25	Double block/bleed Valve required				Yes	☒	No			
26	Interlock systems				Yes	☒	No			
27	Fluid Type				Liquid only		Gas only		Liquid + Gas	☒
28	Orientation				Horizontal	☒	Vertical			
29	Reversible Use									
30	Type				Launcher		Receiver	☒	Launcher & Receiver	
31										
32	Design Factor				0.6					
33	ANSI Pressure Class				300		600		900	☒
34					1500		2500			
35	Design Pressure				110 barg					
36	Design Temperature				100 °C					
37	Ambient Temperature				Minimum	18 °C				
38					Maximum	38 °C				
39	Provision for future hot-tapping/stopping				Yes		No	☒		
40	Fluid Nature				toxic		corrosive		sour	
41					non-toxic		non-corrosive		non-sour	☒
42					☒	In shaded boxes, use a 'Check Mark' to indicate selection				
43										
44	ADDITIONAL DATA (use continuation sheet if required)									
45										
46										
47										
48	INFORMATION TO BE SUBMITTED WITH THE TENDER									
49										
50										
51	NOTES									
52										
53										
54										
55										
R02	08-Dec-22	Issue for Client Review	S.B	O.J	A.D					
R01	28-Jul-22	Issue for Client Review	S.B	O.J	A.D					
Rev.	Date	Issue Description	Originated	Checked	Approved	Client				
 				BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE						
				Project No:		SHEET				
				Datasheet No:		1 of 1				

Process Data Sheet for BBOU Pig Launcher (BBOU-A-200)

1	Tag No. :	BBOU-A-200	Client:	BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD			
2	Service:	GAS	Plant Location:	BBOU			
3	P & ID No. :	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_2	Manufacturer:				
4	Calculation Ref.:	R02	Purchase Order No. :				
5	Line No.:		No. of Units:	ONE			
6	Equipment No.:		Service Description.:				
11							
12	Applicable Piping Class	900#					
13	Design Code	ANSI B31.4		ANSI B31.8	✓	ANSI B31.3	✓
14	Tag Number(s)	BBOU-A-200					
15	Pipeline Diameter	24 inches (24"/28" minor/major barrels)					
16	Type of Pigs	Normal	✓	Spheres	✓	Other (state below)	
17		Intelligent	✓	Foam	✓		
18	Chemical Injection	Yes		No	✓		
19	Thermowell Installation	Yes		No	✓		
20	Valve Type	Globe	✓	Ball	✓		
21	Intended Pigging Frequency	Quarterly					
22	Power-actuated Valves	Yes		No	✓		
23	Pre-commissioning Drying Method	Vacuum		Methanol		N/A	
24	Type of Valve to be used on Mainline Parts	Weld end		Flanged	✓		
25	Double block/bleed Valve required	Yes	✓	No			
26	Interlock systems	Yes	✓	No			
27	Fluid Type	Liquid only		Gas only		Liquid + Gas	✓
28	Orientation	Horizontal	✓	Vertical			
29	Reversible Use						
30	Type	Launcher	✓	Receiver		Launcher & Receiver	
31							
32	Design Factor	0.6					
33	ANSI Pressure Class	300		600		900	✓
34		1500		2500			
35	Design Pressure	110 barg					
36	Design Temperature	100 °C					
37	Ambient Temperature	Minimum	18 °C				
38		Maximum	38 °C				
39	Provision for future hot-tapping/stopping	Yes		No	✓		
40	Fluid Nature	toxic		corrosive		sour	
41		non-toxic		non-corrosive		non-sour	✓
42							
43		✓	In shaded boxes, use a 'Check Mark' to indicate selection				
44							
45	ADDITIONAL DATA (use continuation sheet if required)						
46							
47							
48							
49	INFORMATION TO BE SUBMITTED WITH THE TENDER						
50							
51							
52	NOTES						
53							
54							
55							
	R02	08-Dec-22	Issue for Client Review	S.B	O.J	A.D	
	R01	28-Jul-22	Issued for Client Review	S.B	O.J	A.D	
	Rev.	Date	Issue Description	Originated	Checked	Approved	Client
	 			BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE			
				Project No:		SHEET	
				Datasheet No:		1 of 1	

Process Data Sheet for ELEP Pig Receiver (ELEP-A-100)

1	Tag No. :	ELEP-A-100	Client:	BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD		
2	Service:	GAS	Plant Location:	ELEPA		
3	P & ID No. :	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_4	Manufacturer:			
4	Calculation Ref.:	R02	Purchase Order No. :			
5	Line No.:		No. of Units:	ONE		
6	Equipment No.:		Service Description.:			
11						
12	Applicable Piping Class	900#				
13	Design Code	ANSI B31.4		ANSI B31.8	☒	ANSI B31.3 ☒
14	Tag Number(s)					
15	Pipeline Diameter	36 inches (36"/40" minor/major barrels)				
16	Type of Pigs	Normal ☒		Spheres ☒		Other (state below)
17		Intelligent ☒		Foam ☒		
18	Chemical Injection	Yes		No ☒		
19	Thermowell Installation	Yes		No ☒		
20	Valve Type	Globe ☒		Ball ☒		
21	Intended Pigging Frequency	Quarterly				
22	Power-actuated Valves	Yes		No ☒		
23	Pre-commissioning Drying Method	Vacuum		Methanol		N/A ☒
24	Type of Valve to be used on Mainline Parts	Weld end		Flanged ☒		
25	Double block/bleed Valve required	Yes ☒		No		
26	Interlock systems	Yes ☒		No		
27	Fluid Type	Liquid only		Gas only		Liquid + Gas ☒
28	Orientation	Horizontal ☒		Vertical		
29	Reversible Use					
30	Type	Launcher		Receiver ☒		Launcher & Receiver ☒
31						
32	Design Factor	0.6				
33	ANSI Pressure Class	300		600		900 ☒
34		1500		2500		
35	Design Pressure	110 barg				
36	Design Temperature	100 °C				
37	Ambient Temperature	Minimum	18 °C			
38		Maximum	38 °C			
39	Provision for future hot-tapping/stopping	Yes		No ☒		
40	Fluid Nature	toxic		corrosive		sour ☒
41		non-toxic		non-corrosive		non-sour ☒
42	☒ In shaded boxes, use a 'Check Mark' to indicate selection					
43						
44	ADDITIONAL DATA (use continuation sheet if required)					
45						
46						
47						
48	INFORMATION TO BE SUBMITTED WITH THE TENDER					
49						
50						
51	NOTES					
52						
53						
54						
55						

R02	08-Dec-22	Issue for Client Review	S.B	O.J	A.D	
R01	25-Jul-22	Issued for Client Review	S.B	O.J	A.D	
Rev.	Date	Issue Description	Originated	Checked	Approved	Client





BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD
FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED
INFRASTRUCTURE

Project No:

SHEET

Datasheet No:

1 of 1

Process Data Sheet for ELEP Pig Receiver (ELEP-A-200)

1	Tag No. :	ELEP-A-200		Client:	BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD		
2	Service:	GAS		Plant Location:	ELEPA		
3	P & ID No. :	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_5		Manufacturer:			
4	Calculation Ref.:	R02		Purchase Order No. :			
5	Line No.:			No. of Units:	ONE		
6	Equipment No.:			Service Description.:			
11							
12	Applicable Piping Class	900#					
13	Design Code	ANSI B31.4		ANSI B31.8	✓	ANSI B31.3	✓
14	Tag Number(s)	ELEP-A-200					
15	Pipeline Diameter	24 inches (24"/28" minor/major barrels)					
16	Type of Pigs	Normal	✓	Spheres	✓	Other (state below)	
17		Intelligent	✓	Foam	✓		
18	Chemical Injection	Yes		No	✓		
19	Thermowell Installation	Yes		No	✓		
20	Valve Type	Globe	✓	Ball	✓		
21	Intended Piggings Frequency	Quarterly					
22	Power-actuated Valves	Yes		No	✓		
23	Pre-commissioning Drying Method	Vacuum		Methanol		N/A	
24	Type of Valve to be used on Mainline Parts	Weld end		Flanged	✓		
25	Double block/bleed Valve required	Yes	✓	No			
26	Interlock systems	Yes	✓	No			
27	Fluid Type	Liquid only		Gas only		Liquid + Gas	✓
28	Orientation	Horizontal	✓	Vertical			
29	Reversible Use						
30	Type	Launcher		Receiver	✓	Launcher & Receiver	
31							
32	Design Factor	0.6					
33	ANSI Pressure Class	300		600		900	✓
34		1500		2500			
35	Design Pressure	110 barg					
36	Design Temperature	100 °C					
36	Ambient Temperature	Minimum	18 °C				
37		Maximum	38 °C				
38	Provision for future hot-tapping/stopping	Yes		No	✓		
39	Fluid Nature	toxic		corrosive		sour	
40		non-toxic		non-corrosive		non-sour	✓
41							
42		✓	In shaded boxes, use a 'Check Mark' to indicate selection				
43							
44	ADDITIONAL DATA (use continuation sheet if required)						
45							
46							
47							
48	INFORMATION TO BE SUBMITTED WITH THE TENDER						
49							
50							
51	NOTES						
52							
53							
54							
55							
R02	08-Dec-22	Issue for Client Review	S.B	O.J	A.D		
R01	28-Jul-22	Issue for Client Review	S.B	O.J	A.D		
Rev.	Date	Issue Description	Originated	Checked	Approved	Client	
 				BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE			
				Project No:		SHEET	
				Datasheet No:		1 of 1	

Process Data Sheet for ELEP Pig Launcher (ELEP-A-300)

1	Tag No. :	ELEP-A-300	Client:		BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD	
2	Service:	GAS	Plant Location:		ELEPA	
3	P & ID No. :	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_2	Manufacturer:			
4	Calculation Ref.:	R02	Purchase Order No. :			
5	Line No.:		No. of Units:		ONE	
6	Equipment No.:		Service Description.:			
11						
12	Applicable Piping Class	900#				
13	Design Code	ANSI B31.4		ANSI B31.8	✓	ANSI B31.3
14	Tag Number(s)	ELEP-A-300				
15	Pipeline Diameter	36 inches (36"/40" minor/major barrels)				
16	Type of Pigs	Normal	✓	Spheres	✓	Other (state below)
17		Intelligent	✓	Foam	✓	
18	Chemical Injection	Yes		No	✓	
19	Thermowell Installation	Yes		No	✓	
20	Valve Type	Globe	✓	Ball	✓	
21	Intended Pigging Frequency	Quarterly				
22	Power-actuated Valves	Yes		No	✓	
23	Pre-commissioning Drying Method	Vacuum		Methanol		N/A
24	Type of Valve to be used on Mainline Parts	Weld end		Flanged	✓	
25	Double block/bleed Valve required	Yes	✓	No		
26	Interlock systems	Yes	✓	No		
27	Fluid Type	Liquid only		Gas only		Liquid + Gas
28	Orientation	Horizontal	✓	Vertical		
29	Reversible Use					
30	Type	Launcher	✓	Receiver		Launcher & Receiver
31						
32	Design Factor	0.6				
33	ANSI Pressure Class	300		600		900
34		1500		2500		
35	Design Pressure	110 barg				
36	Design Temperature	100 °C				
37	Ambient Temperature	Minimum	18 °C			
38		Maximum	38 °C			
39	Provision for future hot-tapping/stopping	Yes		No	✓	
40	Fluid Nature	toxic		corrosive		sour
41		non-toxic		non-corrosive		non-sour
42			✓	In shaded boxes, use a 'Check Mark' to indicate selection		
43						
44	ADDITIONAL DATA (use continuation sheet if required)					
45						
46						
47						
48	INFORMATION TO BE SUBMITTED WITH THE TENDER					
49						
50						
51	NOTES					
52						
53						
54						
55						

R02	08-Dec-22	Issue for Client Review	S.B	O.J	A.D	
R01	28-Jul-22	Issue for Client Review	S.B	O.J	A.D	
Rev.	Date	Issue Description	Originated	Checked	Approved	Client
 				BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE		
				Project No:		SHEET
				Datasheet No:		1 of 1

Process Data Sheet for AKAB Pig Receiver (AKAB-A-100)

1	Tag No. :	AKAB-A-100		Client:		BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD	
2	Service:	GAS		Plant Location:		AKABEL	
3	P & ID No. :	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_5		Manufacturer:			
4	Calculation Ref.:	R02		Purchase Order No. :			
5	Line No.:			No. of Units:		ONE	
6	Equipment No.:			Service Description.:			
11							
12	Applicable Piping Class	900#					
13	Design Code	ANSI B31.4		ANSI B31.8	✓	ANSI B31.3	✓
14	Tag Number(s)	AKAB-A-100					
15	Pipeline Diameter	36 inches (36"/40" minor/major barrels)					
16	Type of Pigs	Normal	✓	Spheres	✓	Other (state below)	
17		Intelligent	✓	Foam	✓		
18	Chemical Injection	Yes		No	✓		
19	Thermowell Installation	Yes		No	✓		
20	Valve Type	Globe	✓	Ball	✓		
21	Intended Pigging Frequency	Quarterly					
22	Power-actuated Valves	Yes		No	✓		
23	Pre-commissioning Drying Method	Vacuum		Methanol		N/A	
24	Type of Valve to be used on Mainline Parts	Weld end		Flanged	✓		
25	Double block/bleed Valve required	Yes	✓	No			
26	Interlock systems	Yes	✓	No			
27	Fluid Type	Liquid only		Gas only		Liquid + Gas	✓
28	Orientation	Horizontal	✓	Vertical			
29	Reversible Use						
30	Type	Launcher		Receiver	✓	Launcher & Receiver	
31							
32	Design Factor	0.6					
33	ANSI Pressure Class	300		600		900	✓
34		1500		2500			
35	Design Pressure	110 barg					
36	Design Temperature	100 °C					
37	Ambient Temperature	Minimum	18 °C				
37		Maximum	38 °C				
38	Provision for future hot-tapping/stopping	Yes		No	✓		
39	Fluid Nature	toxic		corrosive		sour	
40		non-toxic		non-corrosive		non-sour	✓
41							
42		✓	In shaded boxes, use a 'Check Mark' to indicate selection				
43							
44	ADDITIONAL DATA (use continuation sheet if required)						
45							
46							
47							
48	INFORMATION TO BE SUBMITTED WITH THE TENDER						
49							
50							
51	NOTES						
52							
53							
54							
55							
R02	08-Dec-22	Issue for Client Review	S.B	O.J	A.D		
R01	28-Jul-22	Issue for Client Review	S.B	O.J	A.D		
Rev.	Date	Issue Description	Originated	Checked	Approved	Client	
 				BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE			
				Project No:		SHEET	
				Datasheet No:		1 of 1	

3.2. Emergence Shutdown Valves and Blowdown Valves Datasheets

Process Data Sheet for BBOU Shutdown Valve-SDV-1001						
1	Tag No. :	SDV-1001	Client:	BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD		
2	Service:	GAS	Plant Location:	BBOU		
3	P & ID No. :	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_1	Manufacturer:			
4	Calculation Ref.:	R02	Purchase Order No. :			
5	Line No. :	BBOU-16"-P-001-E70	Number of Valves:	One		
6	Service Description:					
7	Fluid/Phase:	Gas				
8	DESIGN DATA					
9	Operating Temperature:	Min/Norm/Max	°C	52 (min) / 58 (max)		
11	Operating Pressure:	Min/Norm/Max	Barg	80/100		
13	Maximum Differential Pressure:		Barg			
14	Vol Flow (BBOU Node):	Min/Norm/Max	MMSCFD	331		
15	Line Size (NB):		in	16		
16	ANSI flange rating:		#	900		
17	BODY (OPTIONAL)					
18	Preliminary Valve Size		in	16		
19	Valve Type:	Full bore/ Reduced bore		Full bore		
20	Material of Construction:			CS		
21	Shut Off Class			Tight Shut Off (TSO)		
22	ACTUATOR (OPTIONAL)					
23	Failure Position	Open/Close/Lock		Closed		
24	Closure Time:		sec	12		
25	Actuator Type			Gas-Over-Oil		
26	Position Switches	Open		Open		
27		Shut		Shut		
28	Backup Air Bottle Required:					
29	NOTES: For design temperatures and pressures refer to the line list and valve list.					
30						
45						
46						
47						
48						
49						
50						
R02	08-Dec-22	Issue for Client Review	S.B	K.Y	A.D	
R01	28-Jul-22	Issue for Client Review	S.B	K.Y	A.D	
Rev.	Date	Issue Description	Originated	Checked	Approved	Client
 			BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE			
			Project No:		SHEET	
			Datasheet No:		1 of 1	

Process Data Sheet for BBOU Blowdown Valve-BDV-1001

1	Tag No. :	BDV-1001		Client:	BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD	
2	Service:	GAS		Plant Location:	BBOU	
3	P & ID No. :	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_1		Manufacturer:		
4	Calculation Ref.:	R02		Purchase Order No. :		
5	Line No. :	BBOU-4"-P-004-E23		Number of Valves:	One	
6	Service Description:					
7	Fluid/Phase:	Gas				
8	DESIGN DATA					
9	Operating Temperature:	Min/Norm/Max	°C	52 (min) / 58 (max)		
11	Operating Pressure:	Min/Norm/Max	Barg	80/100		
13	Maximum Differential Pressure:		Barg			
14	Vol Flow (BBOU Node):	Min/Norm/Max	MMSCFD	331		
15	Line Size (NB):		in	4		
16	ANSI flange rating:		#	900		
17	BODY (OPTIONAL)					
18	Preliminary Valve Size		in	4		
19	Valve Type:	Full bore/ Reduced bore		Full bore		
20	Material of Construction:			CS		
21	Shut Off Class			Tight Shut Off (TSO)		
22	ACTUATOR (OPTIONAL)					
23	Failure Position	Open/Close/Lock		Open		
24	Closure Time:		sec	12		
25	Actuator Type			Gas-Over-Oil		
26	Position Switches	Open		Open		
27		Shut		Shut		
28	Backup Air Bottle Required:					
29	NOTES: For design temperatures and pressures refer to the line list.					
30						
31						
46						
47						
48						
49						
50						
R02	08-Dec-22	Issue for Client Review		S.B	K.Y	A.D
R01	28-Jul-22	Issue for Client Review		S.B	K.Y	A.D
Rev.	Date	Issue Description		Originated	Checked	Approved
 				BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE		
				Project No:	SHEET	
				Datasheet No:	1 of 1	

Process Data Sheet for BBOU Shutdown Valve-SDV-1002

1	Tag No. :	SDV-1002	Client:	BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD		
2	Service:	GAS	Plant Location:	BBOU		
3	P & ID No. :	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_1	Manufacturer:			
4	Calculation Ref.:	R02	Purchase Order No. :			
5	Line No. :	BBOU-24"-P-012-CL900	Number of Valves:	One		
6	Service Description:					
7	Fluid/Phase:	Gas				
8	DESIGN DATA					
9	Operating Temperature:	Min/Norm/Max	°C	85 (min) / 84 (max)		
11	Operating Pressure:	Min/Norm/Max	Barg	80/100		
13	Maximum Differential Pressure:		Barg			
14	Vol Flow (Opughe Node):	Min/Norm/Max	MMSCFD	182		
15	Line Size (NB):		in	24		
16	ANSI flange rating:		#	900		
17	BODY (OPTIONAL)					
18	Preliminary Valve Size		in	24		
19	Valve Type:	Full bore/ Reduced bore		Full bore		
20	Material of Construction:			CS		
21	Shut Off Class			Tight Shut Off (TSO)		
22	ACTUATOR (OPTIONAL)					
23	Failure Position	Open/Close/Lock		Closed		
24	Closure Time:		sec	12		
25	Actuator Type			Gas-Over-Oil		
26	Position Switches	Open		Open		
27		Shut		Shut		
28	Backup Air Bottle Required:					
29	NOTES: For design temperatures and pressures refer to the line list.					
30						
31						
32						
33						
34						
35						
36						
37						
38						
39						
40						
41						
42						
43						
44						
45						
46						
47						
48						
49						
50						
R02	08-Dec-22	Issue for Client Review	S.B	K.Y	A.D	
R01	28-Jul-22	Issue for Client Review	S.B	K.Y	A.D	
Rev.	Date	Issue Description	Originated	Checked	Approved	Client
 				BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE		
				Project No:		SHEET
				Datasheet No:		1 of 1

Process Data Sheet for BBOU Shutdown Valve-SDV-1003

1	Tag No. :	SDV-1003	Client:	BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD		
2	Service:	GAS	Plant Location:	BBOU		
3	P & ID No. :	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_2	Manufacturer:			
4	Calculation Ref.:	R01	Purchase Order No. :			
5	Line No. :	BBOU-24"-P-013-CL900	Number of Valves:	One		
6	Service Description:					
7	Fluid/Phase:	Gas				
8	DESIGN DATA					
9	Operating Temperature:	Min/Norm/Max	°C	58 (min) / 62 (max)		
11	Operating Pressure:	Min/Norm/Max	Barg	80/100		
13	Maximum Differential Pressure:		Barg			
14	Vol Flow (BBOU to ELEP Node):	Min/Norm/Max	MMSCFD	331		
15	Line Size (NB):		in	24		
16	ANSI flange rating:		#	900		
17	BODY (OPTIONAL)					
18	Preliminary Valve Size		in	24		
19	Valve Type:	Full bore/ Reduced bore		Full bore		
20	Material of Construction:			CS		
21	Shut Off Class			Tight Shut Off (TSO)		
22	ACTUATOR (OPTIONAL)					
23	Failure Position	Open/Close/Lock		Closed		
24	Closure Time:		sec	12		
25	Actuator Type			Gas-Over-Oil		
26	Position Switches	Open				
27		Shut		Shut		
28	Backup Air Bottle Required:					
29	NOTES: For design temperatures and pressures refer to the line list.					
30						
31						
32						
33						
34						
35						
36						
37						
38						
39						
40						
41						
42						
43						
44						
45						
46						
47						
48						
49						
50						
R02	08-Dec-22	Issue for Client Review	S.B	K.Y	A.D	
R01	28-Jul-22	Issue for Client Review	S.B	K.Y	A.D	
Rev.	Date	Issue Description	Originated	Checked	Approved	Client
 			BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE			
			Project No:		SHEET	
			Datasheet No:		1 of 1	

Process Data Sheet for ELEPA Shutdown Valve-SDV-1001

1	Tag No. :	SDV-1001	Client:	BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD		
2	Service:	GAS	Plant Location:	ELEPA		
3	P & ID No. :	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_4	Manufacturer:			
4	Calculation Ref.:	R02	Purchase Order No. :			
5	Line No. :	ELEP-16"-P-001-E70	Number of Valves:	One		
6	Service Description:					
7	Fluid/Phase:	Gas				
8	DESIGN DATA					
9	Operating Temperature:	Min/Norm/Max	°C	80 (min) / 86 (max)		
11	Operating Pressure:	Min/Norm/Max	Barg	80/100		
13	Maximum Differential Pressure:		Barg			
14	Vol Flow (ELEP Node):	Min/Norm/Max	MMSCFD	180		
15	Line Size (NB):		in	16		
16	ANSI flange rating:		#	900		
17	BODY (OPTIONAL)					
18	Preliminary Valve Size		in	16		
19	Valve Type:	Full bore/ Reduced bore		Full bore		
20	Material of Construction:			CS		
21	Shut Off Class			Tight Shut Off (TSO)		
22	ACTUATOR (OPTIONAL)					
23	Failure Position	Open/Close/Lock		Closed		
24	Closure Time:		sec	12		
25	Actuator Type			Gas-Over-Oil		
26	Position Switches	Open		Open		
27		Shut		Shut		
28	Backup Air Bottle Required:					
29	NOTES: For design temperatures and pressures refer to the line list.					
30						
31						
32						
33						
34						
35						
36						
37						
38						
39						
40						
41						
42						
43						
44						
45						
46						
47						
48						
49						
50						
R02	08-Dec-22	Issue for Client Review	S.B	K.Y	A.D	
R01	28-Jul-22	Issue for Client Review	S.B	K.Y	A.D	
Rev.	Date	Issue Description	Originated	Checked	Approved	Client
 			BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE			
			Project No:		SHEET	
			Datasheet No:		1 of 1	

Process Data Sheet for ELEPA Blowdown Valve-BDV-1001

1	Tag No. :	BDV-1001	Client:	BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD		
2	Service:	GAS	Plant Location:	ELEPA		
3	P & ID No. :	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_4	Manufacturer:			
4	Calculation Ref.:	R02	Purchase Order No. :			
5	Line No. :	ELEP-4"-P-004-E26	Number of Valves:	One		
6	Service Description:					
7	Fluid/Phase:	Gas				
8	DESIGN DATA					
9	Operating Temperature:	Min/Norm/Max	°C	80 (min) / 86 (max)		
11	Operating Pressure:	Min/Norm/Max	Barg	80/100		
13	Maximum Differential Pressure:		Barg			
14	Vol Flow (ELEP Node):	Min/Norm/Max	MMSCFD	180		
15	Line Size (NB):		in	4		
16	ANSI flange rating:		#	900		
17	BODY (OPTIONAL)					
18	Preliminary Valve Size		in	4		
19	Valve Type:	Full bore/ Reduced bore		Full bore		
20	Material of Construction:			CS		
21	Shut Off Class			Tight Shut Off (TSO)		
22	ACTUATOR (OPTIONAL)					
23	Failure Position	Open/Close/Lock		Open		
24	Closure Time:		sec	12		
25	Actuator Type			Gas-Over-Oil		
26	Position Switches	Open		Open		
27		Shut		Shut		
28	Backup Air Bottle Required:					
29	NOTES: For design temperatures and pressures refer to the line list.					
30						
31						
32						
33						
34						
35						
36						
37						
38						
39						
40						
41						
42						
43						
44						
45						
46						
47						
48						
49						
50						
R02	08-Dec-22	Issue for Client Review	S.B	K.Y	A.D	
R01	28-Jul-22	Issue for Client Review	S.B	K.Y	A.D	
Rev.	Date	Issue Description	Originated	Checked	Approved	Client
 			BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE			
			Project No:		SHEET	
			Datasheet No:		1 of 1	

Process Data Sheet for ELEPA Shutdown Valve-SDV-1002

1	Tag No. :	SDV-1002	Client:	BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD		
2	Service:	GAS	Plant Location:	ELEPA		
3	P & ID No. :	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_4	Manufacturer:			
4	Calculation Ref.:	R02	Purchase Order No. :			
5	Line No. :	ELEP-36"-P-014-CL900	Number of Valves:	One		
6	Service Description:					
7	Fluid/Phase:	Gas				
8	DESIGN DATA					
9	Operating Temperature:	Min/Norm/Max	°C	60 (min) / 64 (max)		
11	Operating Pressure:	Min/Norm/Max	Barg	80/100		
13	Maximum Differential Pressure:		Barg			
14	Vol Flow (NUNR Node):	Min/Norm/Max	MMSCFD	386		
15	Line Size (NB):		in	36		
16	ANSI flange rating:		#	900		
17	BODY (OPTIONAL)					
18	Preliminary Valve Size		in	36		
19	Valve Type:	Full bore/ Reduced bore		Full bore		
20	Material of Construction:			CS		
21	Shut Off Class			Tight Shut Off (TSO)		
22	ACTUATOR (OPTIONAL)					
23	Failure Position	Open/Close/Lock		Closed		
24	Closure Time:		sec	12		
25	Actuator Type			Gas-Over-Oil		
26	Position Switches	Open		Open		
27		Shut		Shut		
28	Backup Air Bottle Required:					
29	NOTES: For design temperatures and pressures refer to the line list.					
30						
31						
32						
33						
34						
35						
36						
37						
38						
39						
40						
41						
42						
43						
44						
45						
46						
47						
48						
49						
50						

R02	08-Dec-22	Issue for Client Review	S.B	K.Y	A.D	
R01	28-Jul-22	Issue for Client Review	S.B	K.Y	A.D	
Rev.	Date	Issue Description	Originated	Checked	Approved	Client
 			BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE			
			Project No:		SHEET	
			Datasheet No:		1 of 1	

Process Data Sheet for ELEPA Shutdown Valve-SDV-1003

1	Tag No. : SDV-1003		Client: BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD	
2	Service: GAS		Plant Location: ELEPA	
3	P & ID No. : EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_5		Manufacturer:	
4	Calculation Ref.: R02		Purchase Order No. :	
5	Line No. : ELEP-24"-P-013-CL900		Number of Valves: One	
6	Service Description:			
7	Fluid/Phase: Gas			
8	DESIGN DATA			
9	Operating Temperature:	Min/Norm/Max	°C	58 (min) / 60 (max)
11	Operating Pressure:	Min/Norm/Max	Barg	80/100
13	Maximum Differential Pressure:		Barg	
14	Vol Flow (BBOU - ELEP Node):	Min/Norm/Max	MMSCFD	331
15	Line Size (NB):		in	24
16	ANSI flange rating:		#	900
17	BODY (OPTIONAL)			
18	Preliminary Valve Size		in	24
19	Valve Type:	Full bore/ Reduced bore		Full bore
20	Material of Construction:			CS
21	Shut Off Class			Tight Shut Off (TSO)
22	ACTUATOR (OPTIONAL)			
23	Failure Position	Open/Close/Lock		Closed
24	Closure Time:		sec	12
25	Actuator Type			Gas-Over-Oil
26	Position Switches	Open		Open
27		Shut		Shut
28	Backup Air Bottle Required:			
29	NOTES: For design temperatures and pressures refer to the line list.			
30				
31				
32				
33				
34				
35				
36				
37				
38				
39				
40				
41				
42				
43				
44				
45				
46				
47				
48				
49				
50				

R02	08-Dec-22	Issue for Client Review	S.B	K.Y	A.D	
R01	28-Jul-22	Issue for Client Review	S.B	K.Y	A.D	
Rev.	Date	Issue Description	Originated	Checked	Approved	Client
 			BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE			
			Project No:		SHEET	
			Datasheet No:		1 of 1	

Process Data Sheet for ELEPA Shutdown Valve-SDV-1004

1	Tag No. :	SDV-1004	Client:	BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD		
2	Service:	GAS	Plant Location:	ELEPA		
3	P & ID No. :	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_6	Manufacturer:			
4	Calculation Ref.:	R01	Purchase Order No. :			
5	Line No. :	ELEP-36"-P-011-CL900	Number of Valves:	One		
6	Service Description:					
7	Fluid/Phase:	Gas				
8	DESIGN DATA					
9	Operating Temperature:	Min/Norm/Max	°C	60 (min) / 63 (max)		
11	Operating Pressure:	Min/Norm/Max	Barg	80/100		
13	Maximum Differential Pressure:		Barg			
14	Vol Flow (ELEP - AKABEL Node):	Min/Norm/Max	MMSCFD	386		
15	Line Size (NB):		in	36		
16	ANSI flange rating:		#	900		
17	BODY (OPTIONAL)					
18	Preliminary Valve Size		in	36		
19	Valve Type:	Full bore/ Reduced bore		Full bore		
20	Material of Construction:			CS		
21	Shut Off Class			Tight Shut Off (TSO)		
22	ACTUATOR (OPTIONAL)					
23	Failure Position	Open/Close/Lock		Closed		
24	Closure Time:		sec	12		
25	Actuator Type			Gas-Over-Oil		
26	Position Switches	Open		Open		
27		Shut		Shut		
28	Backup Air Bottle Required:					
29	NOTES: For design temperatures and pressures refer to the line list.					
30						
31						
32						
33						
34						
35						
36						
37						
38						
39						
40						
41						
42						
43						
44						
45						
46						
47						
48						
49						
50						
R02	08-Dec-22	Issue for Client Review	S.B	K.Y	A.D	
R01	28-Jul-22	Issue for Client Review	S.B	K.Y	A.D	
Rev.	Date	Issue Description	Originated	Checked	Approved	Client
 			BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE			
			Project No:		SHEET	
			Datasheet No:		1 of 1	

Process Data Sheet for AKAB Shutdown Valve-SDV-1001

1	Tag No. :	SDV-1001	Client:	BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD		
2	Service:	GAS	Plant Location:	AKABEL		
3	P & ID No. :	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_8	Manufacturer:			
4	Calculation Ref.:	R01	Purchase Order No. :			
5	Line No. :	AKAB-36"-P-015-CL900	Number of Valves:	One		
6	Service Description:					
7	Fluid/Phase:	Gas				
8	DESIGN DATA					
9	Operating Temperature:	Min/Norm/Max	°C	38 (min) / 39 (max)		
11	Operating Pressure:	Min/Norm/Max	Barg	60/65		
13	Maximum Differential Pressure:		Barg			
14	Vol Flow (Max gasflow from ELEP):	Min/Norm/Max	MMSCFD	386		
15	Line Size (NB):		in	36		
16	ANSI flange rating:		#	900		
17	BODY (OPTIONAL)					
18	Preliminary Valve Size		in	36		
19	Valve Type:	Full bore/ Reduced bore		Full bore		
20	Material of Construction:			CS		
21	Shut Off Class			Tight Shut Off (TSO)		
22	ACTUATOR (OPTIONAL)					
23	Failure Position	Open/Close/Lock		Closed		
24	Closure Time:		sec	12		
25	Actuator Type			Gas-Over-Oil		
26	Position Switches	Open		Open		
27		Shut		Shut		
28	Backup Air Bottle Required:					
29	NOTES: For design temperatures and pressures refer to the line list.					
30						
31						
32						
33						
34						
35						
36						
37						
38						
39						
40						
41						
42						
43						
44						
45						
46						
47						
48						
49						
50						
R02	08-Dec-22	Issue for Client Review	S.B	K.Y	A.D	
R01	28-Jul-22	Issue for Client Review	S.B	K.Y	A.D	
Rev.	Date	Issue Description	Originated	Checked	Approved	Client
 			BRASS FERTILIZER AND PETROCHEMICAL COMPANY LTD FEED FOR GAS GATHERING PIPELINE AND ASSOCIATED INFRASTRUCTURE			
			Project No:		SHEET	
			Datasheet No:		1 of 1	

3.3. PSV Datasheets

DATASHEET FOR BBOU PIPELINE PRESSURE SAFETY VALVE (PRV-1001/1002)					
Location: BBOU			Package Item: Pressure Safety Valve		
VALVE ID	Tag No:		PRV-1001 / 1002		
	Service		BBOU to ELEPA Pipeline Over Pressure Protection		
	Location		BBOU to ELEPA Pipeline		
	P&ID No.		EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_1		
	Line No / Vessel No.		BBOU-24"-P-003-C03		
Hazardous Area Classification		Zone 1, Zone 2	Group IIB	T3	
GENERAL	Valve Type		Conventional Spring Type		
	Safety / Relief	Balanced	Relief	Yes	
	Nozzle	Bonnet	Full	Closed	
CONNECTIONS	Inlet	Flange Rating	3"	900#	
	Outlet	Flange Rating	6"	150#	
	Facing		RF (ASME 16.5)		
CONSTRUCTION (MAIN VALVE)	Standard		API 526		
	Body		ASTM A216 WCB / WCC		
	Cap		ASTM A216 WCB / WCC		
	Trim		ASTM A182 F316		
	Seat		Metal to Metal		
	Seals		Elastomeric (ED Certified)		
PILOT VALVE	Body		N/A		
	Trim	Spring	N/A		
	Seat		N/A		
	Seals		N/A		
	Diaphragm		N/A		
SIZING DATA	Design Code		API 520, API 521, API 526		
	Fluid (SG rel to air)		Natural Gas (SG = 0.63767)		
	Sizing Basis		Fire		
	Design/Oper. Temperature (°C)	Design/Oper. Pressure (barg)	-.29/100 / 50	110 barg / 98.0 barg	
	Relieving Temperature (°C)	Relieving Pressure (barg)	95.14	133.1	
	Set Pressure (barg)	CDTP	110		
	% Overpressure		21.00		
	Constant Back Pressure (barg)		0		
	Total Back Pressure (barg)		0		
	Molecular Weight, M		18.46		
	Density kg/m³		90.96		
	Compressibility, Z		0.889		
	Ratio of Sp. Heats, k (Cp / Cv)		1.486		
	Viscosity (cP)		0.017		
	Gas Constant, C		373		
	Kd, Gas		0.975		
	Kb	Kc	1.00	1.00	
SIZING / SELECTION SUMMARY	Calculated Area (in²)		1.344		
	Selected Area (in²)		1.838		
	Orifice Designation		K		
	Actual Relief Capacity, W (kg/h)		65330		
	Rated Flow (kg/h)		89350		
	Estimated Reaction Force		VTA		
	Estimated Noise Level (db)		VTA		
TEST & CERTIFICATES	Material Certificate		Required		
	NDE Test		Required		
PURCHASE	Manufacturer		Pentair or Equivalent		
	Model		VTA		
NOTES					
1) This datasheet shall be read in conjunction with Pressure Safety Valve Specification,					
2) This document is based on the latest revision of PID					
HOLDS					

DATASHEET FOR ELEPA PIPELINE PRESSURE SAFETY VALVE (PRV-1001/1002)					
Location: BBOU			Package Item: Pressure Safety Valve		
VALVE ID	Tag No:		PRV-1001 / 1002		
	Service		ELEPA to AKABEL Pipeline Over Pressure Protection		
	Location		ELEPA to AKABEL Pipeline		
	P&ID No.		EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_4		
	Line No / Vessel No.		ELEP-36"-P-005-C03		
	Harzadous Area Classification		Zone 1, Zone 2	Group IIB	T3
GENERAL	Valve Type		Conventional Spring Type		
	Safety / Relief	Balanced	Relief	Yes	
	Nozzle	Bonnet	Full	Closed	
CONNECTIONS	Inlet	Flange Rating	3"	900#	
	Outlet	Flange Rating	6"	150#	
	Facing		RF (ASME 16.5)		
CONSTRUCTION (MAIN VALVE)	Standard		API 526		
	Body		ASTM A216 WCB / WCC		
	Cap		ASTM A216 WCB / WCC		
	Trim		ASTM A182 F316		
	Seat		Metal to Metal		
	Seals		Elastomeric (ED Certified)		
PILOT VALVE	Body		N/A		
	Trim	Spring	N/A		
	Seat		N/A		
	Seals		N/A		
	Diaphragm		N/A		
SIZING DATA	Design Code		API 520, API 521, API 526		
	Fluid (SG rel to air)		Natural Gas (SG = 0.63767)		
	Sizing Basis		Fire		
	Design/Oper. Temperature (°C)	Design/Oper. Pressure (barg)	-29/100 / 50		110 barg / 98.0 barg
	Relieving Temperature (°C)	Relieving Pressure (barg)	95.14		133.1
	Set Pressure (barg)	CDTP	110		
	% Overpressure		21.00		
	Constant Back Pressure (barg)		0		
	Total Back Pressure (barg)		0		
	Molecular Weight, M		18.46		
	Density kg/m³		90.96		
	Compressibility, Z		0.889		
	Ratio of Sp. Heats, k (Cp / Cv)		1.486		
	Viscosity (cP)		0.017		
	Gas Constant, C		373		
	Kd, Gas		0.975		
	Kb	Kc	1.00		1.00
	SIZING / SELECTION SUMMARY	Calculated Area (in²)		1.344	
Selected Area (in²)		1.838			
Orifice Designation		K			
Actual Relief Capacity, W (kg/h)		65330			
Rated Flow (kg/h)		89350			
Estimated Reaction Force		VTA			
Estimated Noise Level (db)		VTA			
TEST & CERTIFICATES	Material Certificate		Required		
	NDE Test		Required		
PURCHASE	Manufacturer		Pentair or Equivalent		
	Model		VTA		
NOTES					
1) This datasheet shall be read in conjunction with Pressure Safety Valve Specification,					
2) This document is based on the latest revision of PID					
HOLDS					

DATASHEET FOR BBOU PIG RECEIVER PRESSURE SAFETY VALVE (PSV 1004)					
Location: BBOU			Package Item: Pressure Safety Valve		
VALVE ID	Tag No:		PRV-1004		
	Service		BBOU PIG RECEIVER Over Pressure Protection		
	Location		BBOU PIG RECEIVER		
	P&ID No.		EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_1		
	Line No / Vessel No.		BBOU-A-100		
Harzadous Area Classification		Zone 1, Zone 2	Group IIB	T3	
GENERAL	Valve Type		Conventional Spring Type		
	Safety / Relief	Balanced	Relief	Yes	
	Nozzle	Bonnet	Full	Closed	
CONNECTIONS	Inlet	Flange Rating	1.5"	900#	
	Outlet	Flange Rating	2"	300#	
	Facing		RF (ASME 16.5)		
CONSTRUCTION (MAIN VALVE)	Standard		API 526		
	Body		ASTM A216 WCB / WCC		
	Cap		ASTM A216 WCB / WCC		
	Trim		ASTM A182 F316		
	Seat		Metal to Metal		
	Seals		Elastomeric (ED Certified)		
PILOT VALVE	Body		N/A		
	Trim	Spring	N/A		
	Seat		N/A		
	Seals		N/A		
	Diaphragm		N/A		
SIZING DATA	Design Code		API 520, API 521, API 526		
	Fluid (SG rel to air)		Natural Gas (SG = 0.63767)		
	Sizing Basis		Fire		
	Design/Oper. Temperature (°C)	Design/Oper. Pressure (barg)	-29/100 / 50		110 barg / 96 barg
	Relieving Temperature (°C)	Relieving Pressure (barg)	184.2		133.1
	Set Pressure (barg)	CDTP	110		
	% Overpressure		21.00		
	Constant Back Pressure (barg)		0		
	Total Back Pressure (barg)		0		
	Molecular Weight, M		17.89		
	Density kg/m³		139.1		
	Compressibility, Z		0.7217		
	Ratio of Sp. Heats, k (Cp / Cv)		1.988		
	Viscosity (cP)		0.01755		
	Gas Constant, C		373		
	Kd, Gas		0.975		
	Kb	Kc	1.00		1.00
	SIZING / SELECTION SUMMARY	Calculated Area (in²)		0.031	
Selected Area (in²)		0.11			
Orifice Designation		D			
Actual Relief Capacity, W (kg/h)		1501.00			
Rated Flow (kg/h)		5227			
Estimated Reaction Force		VTA			
Estimated Noise Level (db)		VTA			
TEST & CERTIFICATES	Material Certificate		Required		
	NDE Test		Required		
PURCHASE	Manufacturer		Pentair or Equivalent		
	Model		VTA		
NOTES					
1) This datasheet shall be read in conjunction with Pressure Safety Valve Specification,					
2) This document is based on the latest revision of PID					
HOLDS					

DATASHEET FOR BBOU PIG LAUNCHER PRESSURE SAFETY VALVE (PSV 1003)						
Location: BBOU			Package Item: Pressure Safety Valve			
VALVE ID	Tag No:		PRV-1003			
	Service		BBOU PIG LAUNCHER Over Pressure Protection			
	Location		BBOU PIG LAUNCHER			
	P&ID No.		EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_2			
	Line No / Vessel No.		BBOU-A-200			
Harzadous Area Classification		Zone 1, Zone 2	Group IIB	T3		
GENERAL	Valve Type		Conventional Spring Type			
	Safety / Relief	Balanced	Relief	Yes		
	Nozzle	Bonnet	Full	Closed		
CONNECTIONS	Inlet	Flange Rating	1.5"	900#		
	Outlet	Flange Rating	2"	300#		
	Facing		RF (ASME 16.5)			
CONSTRUCTION (MAIN VALVE)	Standard		API 526			
	Body		ASTM A216 WCB / WCC			
	Cap		ASTM A216 WCB / WCC			
	Trim		ASTM A182 F316			
	Seat		Metal to Metal			
	Seals		Elastomeric (ED Certified)			
PILOT VALVE	Body		N/A			
	Trim	Spring	N/A			
	Seat		N/A			
	Seals		N/A			
	Diaphragm		N/A			
SIZING DATA	Design Code		API 520, API 521, API 526			
	Fluid (SG rel to air)		Natural Gas (SG = 0.63767)			
	Sizing Basis		Fire			
	Design/Oper. Temperature (°C)	Design/Oper. Pressure (barg)	-29/100 / 50		110 barg / 98.0 barg	
	Relieving Temperature (°C)	Relieving Pressure (barg)	184.2		133.1	
	Set Pressure (barg)	CDTP	110			
	% Overpressure		21.00			
	Constant Back Pressure (barg)		0			
	Total Back Pressure (barg)		0			
	Molecular Weight, M		17.89			
	Density kg/m³		139.1			
	Compressibility, Z		0.7217			
	Ratio of Sp. Heats, k (Cp / Cv)		1.988			
	Viscosity (cP)		0.01755			
	Gas Constant, C		373			
	SIZING / SELECTION SUMMARY	Kd, Gas		0.975		
		Kb	Kc	1.00	1.00	
Calculated Area (in²)		0.031				
Selected Area (in²)		0.11				
Orifice Designation		D				
Actual Relief Capacity, W (kg/h)		1501.00				
Rated Flow (kg/h)		5227				
Estimated Reaction Force		VTA				
TEST & CERTIFICATES	Material Certificate		Required			
	NDE Test		Required			
PURCHASE	Manufacturer		Pentair or Equivalent			
	Model		VTA			
NOTES						
1) This datasheet shall be read in conjunction with Pressure Safety Valve Specification,						
2) This document is based on the latest revision of PID						
HOLDS						

DATASHEET FOR ELEP PIG RECEIVER PRESSURE SAFETY VALVE (PSV 1004)					
Location: BBOU			Package Item: Pressure Safety Valve		
VALVE ID	Tag No:		PRV-1004		
	Service		ELEP PIG RECEIVER Over Pressure Protection		
	Location		ELEP PIG RECEIVER		
	P&ID No.		EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_4		
	Line No / Vessel No.		ELEP-A-100		
	Harzadous Area Classification		Zone 1, Zone 2	Group IIB	T3
GENERAL	Valve Type		Conventional Spring Type		
	Safety / Relief	Balanced	Relief	Yes	
	Nozzle	Bonnet	Full	Closed	
CONNECTIONS	Inlet	Flange Rating	1.5"	900#	
	Outlet	Flange Rating	2"	300#	
	Facing		RF (ASME 16.5)		
CONSTRUCTION (MAIN VALVE)	Standard		API 526		
	Body		ASTM A216 WCB / WCC		
	Cap		ASTM A216 WCB / WCC		
	Trim		ASTM A182 F316		
	Seat		Metal to Metal		
	Seals		Elastomeric (ED Certified)		
PILOT VALVE	Body		N/A		
	Trim	Spring	N/A		
	Seat		N/A		
	Seals		N/A		
	Diaphragm		N/A		
SIZING DATA	Design Code		API 520, API 521, API 526		
	Fluid (SG rel to air)		Natural Gas (SG = 0.63767)		
	Sizing Basis		Fire		
	Design/Oper. Temperature (°C)	Design/Oper. Pressure (barg)	-.29/100 / 50	110 barg / 96 barg	
	Relieving Temperature (°C)	Relieving Pressure (barg)	195.8	133.1	
	Set Pressure (barg)	CDTP	110		
	% Overpressure		21.00		
	Constant Back Pressure (barg)		0		
	Total Back Pressure (barg)		0		
	Molecular Weight, M		18.44		
	Density kg/m³		65.42		
	Compressibility, Z		0.9695		
	Ratio of Sp. Heats, k (Cp / Cv)		1.311		
	Viscosity (cP)		0.01881		
	Gas Constant, C		373		
	Kd, Gas		0.975		
	Kb	Kc	1.00	1.00	
	SIZING / SELECTION SUMMARY	Calculated Area (in²)		0.0135	
Selected Area (in²)		0.11			
Orifice Designation		D			
Actual Relief Capacity, W (kg/h)		654.90			
Rated Flow (kg/h)		5342			
Estimated Reaction Force		VTA			
Estimated Noise Level (db)		VTA			
TEST & CERTIFICATES	Material Certificate		Required		
	NDE Test		Required		
PURCHASE	Manufacturer		Pentair or Equivalent		
	Model		VTA		
NOTES					
1) This datasheet shall be read in conjunction with Pressure Safety Valve Specification,					
2) This document is based on the latest revision of PID					
HOLDS					

Page 32 of 42

DATASHEET FOR ELEP PIG LAUNCHER PRESSURE SAFETY VALVE (PSV 1005)					
Location: BBOU			Package Item: Pressure Safety Valve		
VALVE ID	Tag No:		PRV-1005		
	Service		ELEP PIG LAUNCHER Over Pressure Protection		
	Location		ELEP PIG LAUNCHER		
	P&ID No.		EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_6		
	Line No / Vessel No.		ELEP-A-300		
	Harzadous Area Classification		Zone 1, Zone 2	Group IIB	T3
GENERAL	Valve Type		Conventional Spring Type		
	Safety / Relief	Balanced	Relief	Yes	
	Nozzle	Bonnet	Full	Closed	
CONNECTIONS	Inlet	Flange Rating	1.5"	900#	
	Outlet	Flange Rating	2"	300#	
	Facing		RF (ASME 16.5)		
CONSTRUCTION (MAIN VALVE)	Standard		API 526		
	Body		ASTM A216 WCB / WCC		
	Cap		ASTM A216 WCB / WCC		
	Trim		ASTM A182 F316		
	Seat		Metal to Metal		
	Seals		Elastomeric (ED Certified)		
PILOT VALVE	Body		N/A		
	Trim	Spring	N/A		
	Seat		N/A		
	Seals		N/A		
	Diaphragm		N/A		
SIZING DATA	Design Code		API 520, API 521, API 526		
	Fluid (SG rel to air)		Natural Gas (SG = 0.63767)		
	Sizing Basis		Fire		
	Design/Oper. Temperature (°C)	Design/Oper. Pressure (barg)	-29/100 / 50	110 barg / 96 barg	
	Relieving Temperature (°C)	Relieving Pressure (barg)	150.6	133.1	
	Set Pressure (barg)	CDTP	110		
	% Overpressure		21.00		
	Constant Back Pressure (barg)		0		
	Total Back Pressure (barg)		0		
	Molecular Weight, M		18.25		
	Density kg/m³		113		
	Compressibility, Z		0.8099		
	Ratio of Sp. Heats, k (Cp / Cv)		1.695		
	Viscosity (cP)		0.01696		
	Gas Constant, C		373		
	Kd, Gas		0.975		
	Kb	Kc	1.00	1.00	
SIZING / SELECTION SUMMARY	Calculated Area (in²)		0.0062		
	Selected Area (in²)		0.11		
	Orifice Designation		D		
	Actual Relief Capacity, W (m³/h)		270.30		
	Rated Flow (m³/h)		4974		
	Estimated Reaction Force		VTA		
	Estimated Noise Level (db)		VTA		
TEST & CERTIFICATES	Material Certificate		Required		
	NDE Test		Required		
PURCHASE	Manufacturer		Pentair or Equivalent		
	Model		VTA		
NOTES					
1) This datasheet shall be read in conjunction with Pressure Safety Valve Specification,					
2) This document is based on the latest revision of PID					
HOLDS					

Page 34 of 42

3.4. RO Datasheets

DATASHEET FOR BBOU PIPELINE RO 1001 RESTRICTION ORIFICE PLATE						
Location: BBOU to ELEPA Pipeline				Item: Paddle Type Concentric Square Edge Restriction Orifice Plate		
GENERAL						
1	Tag No.	P&ID No:	RO 1001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_1		
2	Location		BBOU to ELEPA Pipeline			
3	Service		Depressuring BBOU to ELEPA Pipeline			
4	Line No.	Pipe Class	BBOU-4"-P-004-E23		900#	
5	Line size		4"			
6	Certificate		EN 10204 type 3.1			
7	Hazardous Area Class.	Gas Group	Temp. Class	Zone 1, Zone 2	Group II	T3
8	Ambient temperature (°C)	Min.	Max.	18		38
PROCESS DATA						
9	Process Fluid		Gas			
10	Design / Operating Pressure	Design / Operating Temp (Min/Max)		110 / 80		110 / 100 barg
11	Max DP Across Restriction Orifice Plate (bar)		100			
ORIFICE PLATE						
12	Design Code		API 520, ISO 5167			
13	Type		Paddle Type Concentric Square Edged Restriction Orifice Plate			
14	Material		316 SS			
15	Outer Diameter (in)		4			
16	Thickness (in)		VTA (Note 1)			
17	Bore Diameter (in)		2.24			
18	Beta Ratio		0.57			
19	Flow Direction		Required (etched on the plate)			
20	Nameplate		Required (etched on the plate)			
OTHERS						
21	Manufacturer		VTA			
22	Model		VTA			
23	Purchase Order Number		VTA			
NOTES						
1) Vendor shall advise on the plate thickness with respect to the maximum differential pressure (pressure drop) across the restriction orifice plate.						
HOLDS						
1) This document shall be read in conjunction with the Depressurization Report -						
ABBREVIATIONS						
N/A - Not Applicable						
VTA - Vendor To Advise						

DATASHEET FOR BBOU PIG RECEIVER (BBOU-A-100) RO 1002 RESTRICTION ORIFICE PLATE						
Location: BBOU-A-100 (BBOU PIG RECEIVER)				Item: Paddle Type Concentric Square Edge Restriction Orifice Plate		
GENERAL						
1	Tag No.	P&ID No:	RO 1002	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_1		
2	Location		BBOU-A-100 (BBOU PIG RECEIVER)			
3	Service		Depressuring BBOU-A-100 (BBOU PIG RECEIVER)			
4	Line No.	Pipe Class	BBOU-4"-P-016-E03		900#	
5	Line size		4"			
6	Certificate		EN 10204 type 3.1			
7	Hazardous Area Class.	Gas Group	Temp. Class	Zone 1, Zone 2	Group II	T3
8	Ambient temperature (°C)	Min.	Max.	18		38
PROCESS DATA						
9	Process Fluid		Gas			
10	Design / Operating Pressure	Design / Operating Temp (Min/Max)		110 / 80		110 / 100 barg
11	Max DP Across Restriction Orifice Plate (bar)		100			
ORIFICE PLATE						
12	Design Code		API 520, ISO 5167			
13	Type		Paddle Type Concentric Square Edged Restriction Orifice Plate			
14	Material		316 SS			
15	Outer Diameter (in)		4			
16	Thickness (in)		VTA (Note 1)			
17	Bore Diameter (in)		0.19			
18	Beta Ratio		0.05			
19	Flow Direction		Required (etched on the plate)			
20	Nameplate		Required (etched on the plate)			
OTHERS						
21	Manufacturer		VTA			
22	Model		VTA			
23	Purchase Order Number		VTA			
NOTES						
1) Vendor shall advise on the plate thickness with respect to the maximum differential pressure (pressure drop) across the restriction orifice plate.						
HOLDS						
1) This document shall be read in conjunction with the Depressurization Report -						
ABBREVIATIONS						
N/A - Not Applicable						
VTA - Vendor To Advise						

DATASHEET FOR BBOU PIG LAUNCHER (BBOU-A-200) RO 1003 RESTRICTION ORIFICE PLATE						
Location: BBOU-A-100 (BBOU PIG RECEIVER)				Item: Paddle Type Concentric Square Edge Restriction Orifice Plate		
GENERAL						
1	Tag No.	P&ID No:	RO 1003	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_2		
2	Location		BBOU-A-200 (BBOU PIG LAUNCHER)			
3	Service		Depressuring BBOU-A-200 (BBOU PIG LAUNCHER)			
4	Line No.	Pipe Class	BBOU-6"-P-011-E03		900#	
5	Line size		6"			
6	Certificate		EN 10204 type 3.1			
7	Hazardous Area Class.	Gas Group	Temp. Class	Zone 1, Zone 2	Group II	T3
8	Ambient temperature (°C)	Min.	Max.	18		38
PROCESS DATA						
9	Process Fluid		Gas			
10	Design / Operating Pressure	Design / Operating Temp (Min/Max)		110 / 80		110 / 100 barg
11	Max DP Across Restriction Orifice Plate (bar)		99.3			
ORIFICE PLATE						
12	Design Code		API 520, ISO 5167			
13	Type		Paddle Type Concentric Square Edged Restriction Orifice Plate			
14	Material		316 SS			
15	Outer Diameter (in)		4			
16	Thickness (in)		VTA (Note 1)			
17	Bore Diameter (in)		0.14			
18	Beta Ratio		0.07			
19	Flow Direction		Required (etched on the plate)			
20	Nameplate		Required (etched on the plate)			
OTHERS						
21	Manufacturer		VTA			
22	Model		VTA			
23	Purchase Order Number		VTA			
NOTES						
1) Vendor shall advise on the plate thickness with respect to the maximum differential pressure (pressure drop) across the restriction orifice plate.						
HOLDS						
1) This document shall be read in conjunction with the Depressurization Report -						
ABBREVIATIONS						
N/A - Not Applicable						
VTA - Vendor To Advise						

DATASHEET FOR ELEPA PIPELINE RO 1001 RESTRICTION ORIFICE PLATE						
Location: ELEPA to AKABEL Pipeline				Item: Paddle Type Concentric Square Edge Restriction Orifice Plate		
GENERAL						
1	Tag No.	P&ID No:	RO 1001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_4		
2	Location		ELEPA to AKABEL Pipeline			
3	Service		Depressuring ELEPA to AKABEL Pipeline			
4	Line No.	Pipe Class	ELEP-16"-P-003-E06		900#	
5	Line size		16"			
6	Certificate		EN 10204 type 3.1			
7	Hazardous Area Class.	Gas Group	Temp. Class	Zone 1, Zone 2	Group II	T3
8	Ambient temperature (°C)	Min.	Max.	18		38
PROCESS DATA						
9	Process Fluid		Gas			
10	Design / Operating Pressure	Design / Operating Temp (Min/Max)		110 / 80		110 / 100 barg
11	Max DP Across Restriction Orifice Plate (bar)		98.3			
ORIFICE PLATE						
12	Design Code		API 520, ISO 5167			
13	Type		Paddle Type Concentric Square Edged Restriction Orifice Plate			
14	Material		316 SS			
15	Outer Diameter (in / mm)		4			
16	Thickness (in)		VTA (Note 1)			
17	Bore Diameter (in)		5.68			
18	Beta Ratio		0.73			
19	Flow Direction		Required (etched on the plate)			
20	Nameplate		Required (etched on the plate)			
OTHERS						
21	Manufacturer		VTA			
22	Model		VTA			
23	Purchase Order Number		VTA			
NOTES						
1) Vendor shall advise on the plate thickness with respect to the maximum differential pressure (pressure drop) across the restriction orifice plate.						
HOLDS						
1) This document shall be read in conjunction with the Depressurization Report -						
ABBREVIATIONS						
N/A - Not Applicable						
VTA - Vendor To Advise						

DATASHEET FOR ELEPA PIG RECEIVER (ELEPA-A-100) RO 1002 RESTRICTION ORIFICE PLATE						
Location: ELEPA-A-100 (ELEPA PIG RECEIVER)				Item: Paddle Type Concentric Square Edge Restriction Orifice Plate		
GENERAL						
1	Tag No.	P&ID No:	RO 1002	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_4		
2	Location		ELEPA-A-100 (ELEPA PIG LAUNCHER)			
3	Service		Depressuring ELEPA-A-100 (ELEPA PIG LAUNCHER)			
4	Line No.	Pipe Class	ELEPA-4"-P-016-E06		900#	
5	Line size		4"			
6	Certificate		EN 10204 type 3.1			
7	Hazardous Area Class.	Gas Group	Temp. Class	Zone 1, Zone 2	Group II	T3
8	Ambient temperature (°C)	Min.	Max.	18		38
PROCESS DATA						
9	Process Fluid		Gas			
10	Design / Operating Pressure	Design / Operating Temp (Min/Max)		110 / 80		110 / 100 barg
11	Max DP Across Restriction Orifice Plate (bar)		99.3			
ORIFICE PLATE						
12	Design Code		API 520, ISO 5167			
13	Type		Paddle Type Concentric Square Edged Restriction Orifice Plate			
14	Material		316 SS			
15	Outer Diameter (in)		4			
16	Thickness (in)		VTA (Note 1)			
17	Bore Diameter (in)		0.32			
18	Beta Ratio		0.08			
19	Flow Direction		Required (etched on the plate)			
20	Nameplate		Required (etched on the plate)			
OTHERS						
21	Manufacturer		VTA			
22	Model		VTA			
23	Purchase Order Number		VTA			
NOTES						
1) Vendor shall advise on the plate thickness with respect to the maximum differential pressure (pressure drop) across the restriction orifice plate.						
HOLDS						
1) This document shall be read in conjunction with the Depressurization Report -						
ABBREVIATIONS						
N/A - Not Applicable						
VTA - Vendor To Advise						

DATASHEET FOR ELEPA PIG RECEIVER (ELEP-A-200) RO 1003 RESTRICTION ORIFICE PLATE						
Location: ELEP-A-200 (ELEPA PIG RECEIVER)				Item: Paddle Type Concentric Square Edge Restriction Orifice Plate		
GENERAL						
1	Tag No.	P&ID No:	RO 1003	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_5		
2	Location		ELEP-A-200 (ELEP PIG RECEIVER)			
3	Service		Depressuring ELEP-A-200 (ELEP PIG RECEIVER)			
4	Line No.	Pipe Class	ELEP-4"-P-016-E06		900#	
5	Line size		4"			
6	Certificate		EN 10204 type 3.1			
7	Hazardous Area Class.	Gas Group	Temp. Class	Zone 1, Zone 2	Group II	T3
8	Ambient temperature (°C)	Min.	Max.	18		38
PROCESS DATA						
9	Process Fluid		Gas			
10	Design / Operating Pressure	Design / Operating Temp (Min/Max)		110 / 80		110 / 100 barg
11	Max DP Across Restriction Orifice Plate (bar)		99.3			
ORIFICE PLATE						
12	Design Code		API 520, ISO 5167			
13	Type		Paddle Type Concentric Square Edged Restriction Orifice Plate			
14	Material		316 SS			
15	Outer Diameter (in / mm)		4			
16	Thickness (in)		VTA (Note 1)			
17	Bore Diameter (in)		1.09			
18	Beta Ratio		0.27			
19	Flow Direction		Required (etched on the plate)			
20	Nameplate		Required (etched on the plate)			
OTHERS						
21	Manufacturer		VTA			
22	Model		VTA			
23	Purchase Order Number		VTA			
NOTES						
1) Vendor shall advise on the plate thickness with respect to the maximum differential pressure (pressure drop) across the restriction orifice plate.						
HOLDS						
1) This document shall be read in conjunction with the Depressurization Report -						
ABBREVIATIONS						
N/A - Not Applicable						
VTA - Vendor To Advise						

DATASHEET FOR ELEPA PIG LAUNCHER (ELEP-A-300) RO 1005 RESTRICTION ORIFICE PLATE						
Location: ELEP-A-100 (ELEP PIG RECEIVER)				Item: Paddle Type Concentric Square Edge Restriction Orifice Plate		
GENERAL						
1	Tag No.	P&ID No:	RO 1005	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_6		
2	Location		ELEP-A-300 (ELEPA PIG LAUNCHER)			
3	Service		Depressuring ELEP-A-300 (ELEPA PIG LAUNCHER)			
4	Line No.	Pipe Class	ELEP-6"-P-021-E06		900#	
5	Line size		6"			
6	Certificate		EN 10204 type 3.1			
7	Hazardous Area Class.	Gas Group	Temp. Class	Zone 1, Zone 2	Group II	T3
8	Ambient temperature (°C)	Min.	Max.	18		38
PROCESS DATA						
9	Process Fluid		Gas			
10	Design / Operating Pressure	Design / Operating Temp (Min/Max)		110 / 80		110 / 100 barg
11	Max DP Across Restriction Orifice Plate (bar)		99.3			
ORIFICE PLATE						
12	Design Code		API 520, ISO 5167			
13	Type		Paddle Type Concentric Square Edged Restriction Orifice Plate			
14	Material		316 SS			
15	Outer Diameter (in / mm)		4			
16	Thickness (in)		VTA (Note 1)			
17	Bore Diameter (in)		0.24			
18	Beta Ratio		0.06			
19	Flow Direction		Required (etched on the plate)			
20	Nameplate		Required (etched on the plate)			
OTHERS						
21	Manufacturer		VTA			
22	Model		VTA			
23	Purchase Order Number		VTA			
NOTES						
1) Vendor shall advise on the plate thickness with respect to the maximum differential pressure (pressure drop) across the restriction orifice plate.						
HOLDS						
1) This document shall be read in conjunction with the Depressurization Report -						
ABBREVIATIONS						
N/A - Not Applicable						
VTA - Vendor To Advise						

DATASHEET FOR AKABEL PIG RECEIVER RO 1003 RESTRICTION ORIFICE PLATE						
Location: AKAB-A-200 (AKABA PIG RECEIVER)				Item: Paddle Type Concentric Square Edge Restriction Orifice Plate		
GENERAL						
1	Tag No.	P&ID No:	RO 1003	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_8		
2	Location		AKAB-A-100 (AKAB PIG RECEIVER)			
3	Service		Depressuring AKAB-A-100 (AKAB PIG RECEIVER)			
4	Line No.	Pipe Class	AKAB-4"-P-016-E06		900#	
5	Line size		4"			
6	Certificate		EN 10204 type 3.1			
7	Hazardous Area Class.	Gas Group	Temp. Class	Zone 1, Zone 2	Group II	T3
8	Ambient temperature (°C)	Min.	Max.	18		38
PROCESS DATA						
9	Process Fluid		Gas			
10	Design / Operating Pressure	Design / Operating Temp (Min/Max)		110 / 80		110 / 100 barg
11	Max DP Across Restriction Orifice Plate (bar)		95.9			
ORIFICE PLATE						
12	Design Code		API 520, ISO 5167			
13	Type		Paddle Type Concentric Square Edged Restriction Orifice Plate			
14	Material		316 SS			
15	Outer Diameter (in / mm)		4			
16	Thickness (in)		VTA (Note 1)			
17	Bore Diameter (in)		0.32			
18	Beta Ratio		0.08			
19	Flow Direction		Required (etched on the plate)			
20	Nameplate		Required (etched on the plate)			
OTHERS						
21	Manufacturer		VTA			
22	Model		VTA			
23	Purchase Order Number		VTA			
NOTES						
1) Vendor shall advise on the plate thickness with respect to the maximum differential pressure (pressure drop) across the restriction orifice plate.						
HOLDS						
1) This document shall be read in conjunction with the Depressurization Report -						
ABBREVIATIONS						
N/A - Not Applicable						
VTA - Vendor To Advise						